Aman Raghav

Meerut, UttarPradesh   aman.iimtuniversity@gmail.com   9368057261   amanraghav.com
amanraghav   amanraghavkumar

ABOUT ME!

I am an aspiring Machine Learning Engineer with a strong passion for data-driven problem solving. I've built several end-to-end ML and deep learning projects involving data preprocessing, model training, evaluation, and deployment using tools like Python, TensorFlow, and scikit-learn.

I'm currently looking for opportunities where I can apply my knowledge and grow in real-world ML/AI environments.

Education

IIMT University *Sept 2022 – June 2026*
Bachelor of Technology - Computer Science & Engineering

- **Specialization:** Specializing in Artificial Intelligence and Machine Learning by Samatrix.io (samatrix)

Certifications

- Python For Data Science (NPTEL) *Jul 2023 - Aug 2023*
- Machine Learning (NPTEL) *Jul 2024 - Oct 2024*
- Data Base Management System (NPTEL) *Jan 2025 - Mar 2025*
- Advanced R Programming for Data Analytics in Business (NPTEL) *Jul 2025 - Oct 2025*
- Machine Learning and Pattern Recognition (Samatrix) *01/04/2025*
- Advance Machine Learning (Samatrix) *01/04/2025*
- Deep Learning (Samatrix) *01/04/2025*
- DATA SCIENCE MASTERS (PW Skills) *24/02/2024*
- Generative AI (iNeuron) *03/04/2024*

Skills

- **Programming Languages:** Python
- **Machine Learning:** NumPy, Pandas, Matplotlib, Seaborn,Regression and so more algo. Scikit-learn
- **Deep Learning:** ANN, CNN, RNN, NLP, OpenCV, TensorFlow/Keras
- **Generative AI & LLMs:** Large Language Models (LLMs), LangChain, Hugging Face
- **Tools & Platforms:** SQL, Git, GitHub, Jupyter Notebook, Google Colab, Anaconda, VS Code

Projects

Real Vs Fake Image Detector *github* | Demo

- Developed an AI/ML-based system to classify images as real or fake using deep learning techniques
- Implemented end-to-end workflow including data collection, preprocessing, augmentation, model training, and evaluation using deep learning techniques
- Tools & Technologies: Python, TensorFlow/Keras, NumPy, Pandas, Matplotlib.

- Achieved **98% Accuracy** on training data and Achieved **96% Accuracy** on test data .

AI-Based Image Generator using Pretrained Model *github*

- Built a text-to-image generation system leveraging a pretrained Stable Diffusion model to generate high-quality images from natural language prompts.
- Integrated Hugging Face Transformers and Diffusers library to load and use Stable Diffusion.
- Implemented prompt-based image generation using minimal compute (on CPU/GPU).
- Designed a user-friendly frontend using Streamlit for real-time image generation via text input.
- Optimized model performance for faster rendering and GPU acceleration .

Deep-Chest: Attention-Based Thoracic Disease Classifier *github* | Demo

- Developing a PyTorch-based Deep Learning system to classify 5 chest pathologies from 21,000+ X-ray images.
- Integrated a DenseNet-121 backbone with a custom Spatial Attention Mechanism to improve feature localization in radiological scans.
- Implementing AdamW optimizer and ReduceLROnPlateau scheduler to handle complex medical patterns and prevent overfitting.
- Focusing on AUC-ROC and F1-Score to ensure high diagnostic sensitivity, outperforming standard CNN baselines.

Cost-Efficient RAG Application *github*

- Developed a RAG application using ChromaDB as a local vector database.
- Implemented document ingestion for PDF, HTML, and Markdown files with text chunking.
- Generated embeddings using all-MiniLM-L6-v2 and performed top-k similarity retrieval.
- Built FastAPI APIs for document ingestion and question answering.
- Evaluated retrieval performance using Recall, MRR, Hit Rate, and nDCG metrics.